CSE322: Software Engineering

Lecture Topic: Acceptance Testing & AI Integration

1. Introduction to Acceptance Testing

Acceptance Testing (or User Acceptance Testing - UAT) is the final phase of software testing where the system is tested for acceptability. The purpose is to evaluate the system's compliance with business requirements and verify if it is ready for delivery.

Key Characteristics:

- **Focus:** User needs and business requirements rather than technical implementation.
- **Standard Format:** Often written in *Gherkin Syntax* (Given, When, Then).
- **Outcome:** Binary (Pass/Fail) based on specific criteria.

2. Case Study: "CampusSwap" Marketplace

Scenario: The university is launching a P2P app for students to trade items. You are the QA Lead.

Requirement: "As a student seller, I want to list an item with a photo and price so others can buy it."

3. Quality Spectrum: Good vs. Bad Tests

Bad Acceptance Test	Good Acceptance Test
Scenario: Listing an item. *Given:* I am on the sell page.	**Scenario:** Successfully listing a textbook.

When: I enter the info and a photo. *Then:* The item is posted successfully. **Critique:** Vague. What info? What format? No verifiable success criteria.	*Given:* I am logged in on the "Sell" page. *When:* I upload "book.jpg", enter title "Calculus", and price "$30". *And:* I click "Post Item". *Then:* I see a "Success" notification. *And:* The item appears in my "Active Listings" list. **Critique:** Precise, data-driven, and easily automated.

4. AI/LLM Integration for Quality Control

Artificial Intelligence should be used as a **Collaborative Partner** in the QA process. Students are encouraged to use LLMs for the following activities:

A. The Ambiguity Audit

Use AI to identify subjective language in your requirements.

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Prompt for Student:
"I am a student. Review this test case for ambiguity: 'Then the page should load fast.' Suggest specific metrics."
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B. Edge Case Discovery

AI can simulate 'adversarial' thinking to find what might go wrong.

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Prompt for Student:
"Act as a malicious user on CampusSwap. What are 3 negative test cases for the item listing form?"
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C. Conversion to Automation

Bridge the gap between business requirements and code.

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Prompt for Student:
"Convert my Gherkin scenario [Insert Scenario] into a
Playwright test script using JavaScript."
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5. Grading Rubric

- **Foundational (C):** Correct Gherkin syntax usage.
- **Analytical (B):** Inclusion of negative paths and boundary value analysis.
- **Advanced (A):** Evidence of using AI tools to refine requirements and generate automated scripts.